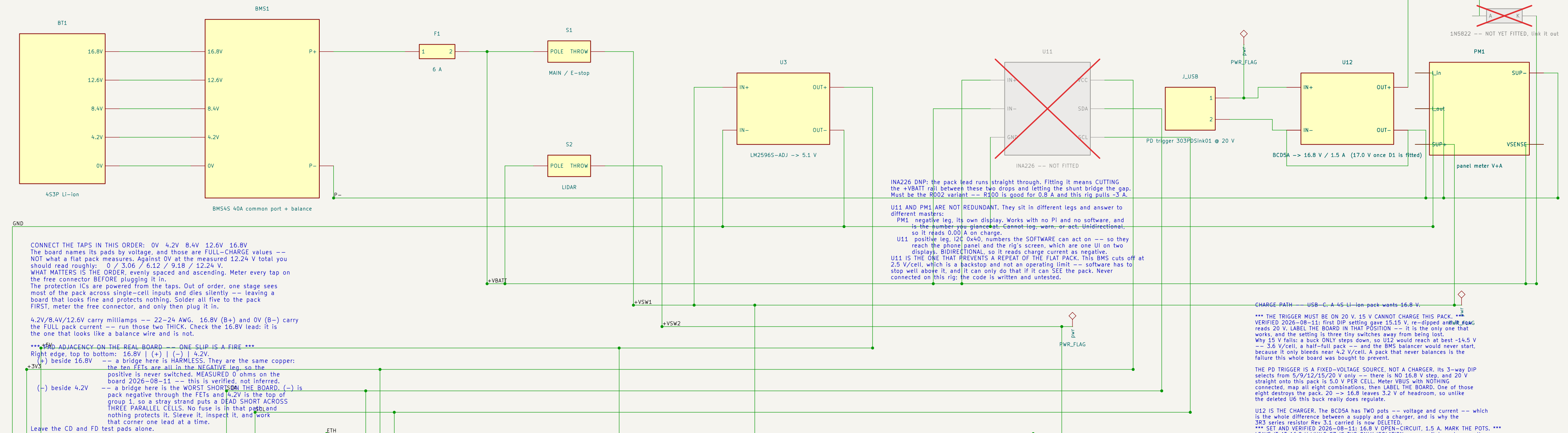
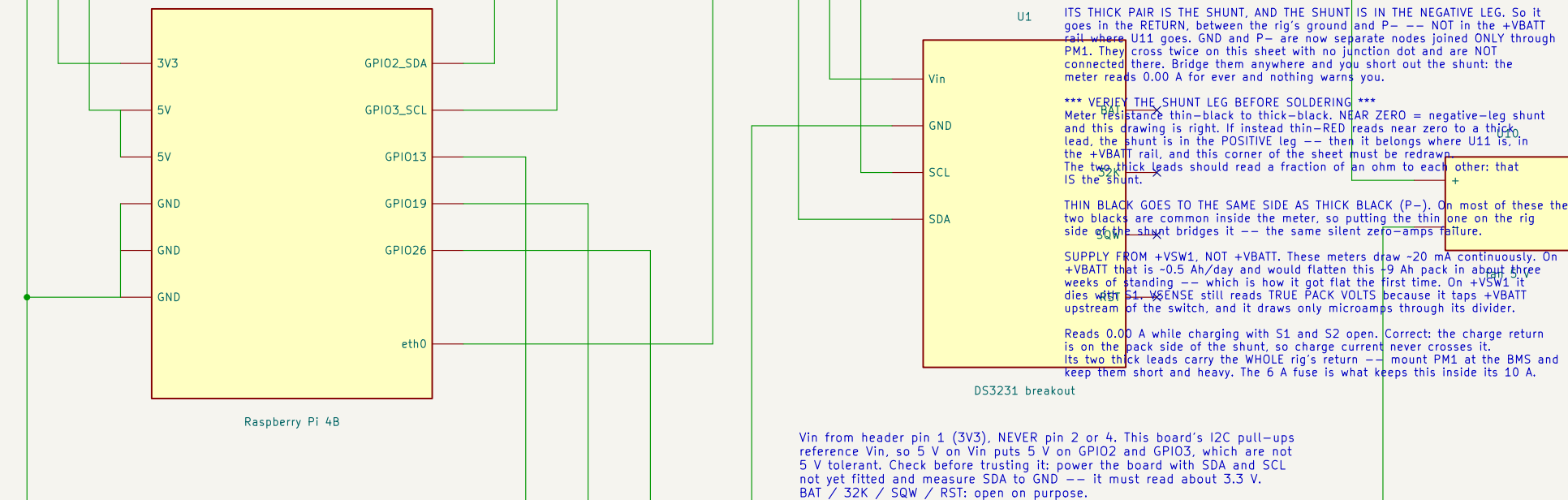


ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge



ZONE B THE PI AND ITS BUS



ZONE C MOTOR CHAIN

NO, THE MOTOR DOES NOT NEED A 12 V BUCK. U6 STAYS DELETED.
A stepper's "12 V" rating is just $I_{rated} \times R_{phase}$ -- the volts you would need with NO chopping. U4 is a current-CHOPPING driver: it PWMs the supply to hold the coil at whatever CUR ADJ is set to, so the supply sees how FAST current rises, not how MUCH. More volts is more torque at speed, and the current limit is what protects the motor. 16.8 V is inside U4's 8-35 V.

*** BUT THE MOTOR HAS ONLY EVER RUN ON A FLAT PACK. ***
At 12.2 V the driver may never have reached its setpoint at speed. At 16.8 V it will. Expect MORE TORQUE AND A HOTTER MOTOR on the first charged run even though nothing was adjusted. CHECK THE MOTOR TEMPERATURE, and run it uncoupled from the head first. Do NOT touch CUR ADJ PWR to compensate.

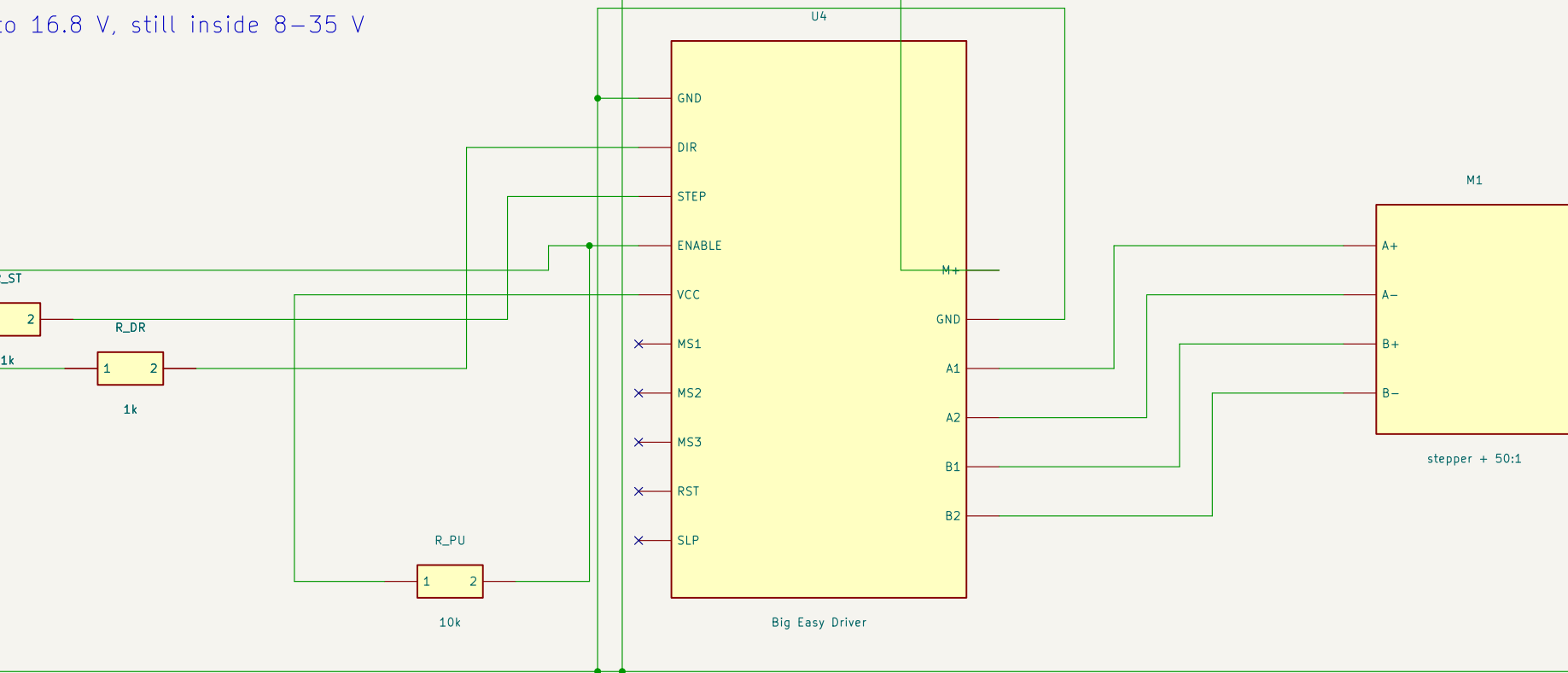
Re-adding U6 would now half-work, which is worse than not working: at 45 it regulates above -13.5 V and drops out below, so the rig would behave one way on a full pack and another on a low one. It would also throw away the torque, add heat and a failure point, and put a -2-3 A module ceiling in front of a chain that draws peaks. The 16.8 V worry belongs to the VLP-16 on S2, not here

R_PU is the gating item -- nothing turns until it is fitted. Every PI GPIO floats as an input for the ~30 s the PI takes to boot, and ENABLE is active low. When this the driver can sit enisured through the time of boot with nothing in the driver of it. It goes on the DRIVER pin, R_EN and pulls up to the driver's own VCC. The PI's 3V3 -- so it still holds with the PI unplugged. Set the APWR jumper to 3.3 V and MEASURE VCC first. MS1-3 open = 1/16 step, via the driver's own pull-ups. RST and SLP are tied together on-board on a genuine PI Easy Driver -- meter them and confirm

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Vin from header pin 1 (3V3). NEVER pin 2 or 4. This board's I2C pull-ups reference Vin, so 5 V on Vin puts 5 V on GPIO2 and GPIO3, which are not 5 V tolerant. Check before trusting it: power the board with SDA and SCL not yet fitted and measure SDA to GND -- it must read about 3.3 V. BAT / 32K / SQW / RST: open on purpose.

to 16.8 V, still inside 8–35 V



ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

- 0 DONE 2026-08-11 ~~~ U12 set on the bench to 16.8 V open-circuit and 1.5 A
- 1 NEITHER POT MAY BE TOUCHED AGAIN: mark them. With the trigger on 20 V
- 2 there is a supply on this bench that would put out 5.0 V/cell on the pack.
- 3 This is the only thing that can be done at the moment. The output of the
- 4 EXPECT THE OUTPUT TO READ -12~-13 V. NOT 16.8 V. THE MOMENT THE FLAT PACK
- 5 IS CONNECTED, THAT IS CC mode holding 1.5 A at whatever the pack sits at
- 6 At 16.8 V as the charger works, it will hold 1.5 A and the pack voltage
- 7 voltage instead of 16.8 V is the charger working, not a lost setting.
- 8 1. CHARGE AT 16.8 V. At 12.22 V this pack sits at 3.05 V/cell and the BMS
- 9 is correctly latched on under-voltage; a common-port board releases when it
- 10 charges. A charge voltage of 16.8 V is needed to get the pack up to 16.8 V.
- 11 2. Four groups ~~~ 0V/4.2V, 4.2V/8.4V, 8.4V/12.6V, 12.6V/16.8V ~~~ each -6.2 V
- 12 when full and ~50 mV of the others. The WEAK GROUP SHOWS UP here
- 13 and the one that trips first is the one that has the lowest threshold. It cost
- 14 16.8 V; the balancer only works at the top, and that hour is the whole
- 15 AT 16.8 V, the balancer is not working. B1/B2 the other. Cross them and the
- 16 motor buzzes, sparks and smells like burnt wire. The motor fan gear has the floor
- 17 plate on it. The motor is not running. The motor is not running. The motor
- 18 with a battery that is flat. The motor is not running. The motor is not running
- 19 started this investigation belonged to the OLD board's higher threshold.
- 20 3. It took about 15.6 V to get the motor to run. The motor is not running.
- 21 4. Only then close S1, and only with the motor uncoupled from the head,

TLS Pie -- Rev 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev 3.0 assumed a 35 pack and was WRONG.
 Rev 3.2 fits the bought parts: BM545 (COMMON PORT) -- no C-pad, so the C-pad is not connected, and the chip is not the star point and the BCD54 CC/CV buffer (two pots, so the 3R3 series resistor is deleted).
 Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection.
 GPIO assignment is owned by J1s_stereoemv

GPIO assignment is owned by `tls_stepper.py`.

Generated by kicad/make_kicad_schematic.

Pack is 4S3P. Star point P-. Common port
Draw 3.2 file the ports height: RMS/6. same

Kizim Robotics

Sheet: /
File: T:\C\B\H\A\A\A

Title: TIS Pie -- Rev 3.2 wiring

Size: A2 Date: 2026-08-11

KiCad E.D.A. 10.0.5

9 10

Title: TLS Pie -- Rev 3.2 wiring schematic

Size: A2	Date: 2026-08-11	Rev: 3
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KiCad E.D.A. 10.0.5		Id: 1/1	
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